

Precipitation and cloud microstructure variations between two Southern Indian stations

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ABSTRACT

Raindrop size distribution (RSD) characteristic variations between two southern Indian stations [Gadanki (13.5° N, 79.2° E) Kadapa (14.47° N, 78.82° E)] using ground based parsivel disdrometer data are studied. Number concentration of mid and large drops is more over Gadanki when compared to Kadapa precipitation. The mean value of mass weighted mean diameter (D_m) is higher in Gadanki than Kadapa precipitation. Both monthly and diurnal variations of D_m show higher values of D_m over Gadanki than Kadapa. After classifying the precipitations systems into stratiform and convective, Gadanki has higher (lower) D_m than Kadapa in stratiform (convective).

Keywords: Raindrop size distribution, mass weighted mean diameter (D_m), cloud effective radius

1. Introduction

Information of raindrop size distribution (RSD) and its variability is important for understanding the processes [1] associated with precipitation and their estimation [2] and for improving the microphysical parameterizations in modeling studies [3]. The RSD is highly variable in space and time. Studying of spatial variability of RSD can provide better understanding of different microphysical process leading to differences in the observed ground rainfall. Hence for the present study an attempt has been made to understand the differences in RSD between two southern Indian stations.

2. Data and Methodology

The PARSIVEL disdrometer is a laser based instrument designed to count and measure simultaneously the size and fall speed of precipitation particles. In brief, parsivel measures particles with diameters between 0.2 and 25 mm and fall velocities between 0.2 and 20 m/s. The core element of the instrument is an optical sensor that produces a horizontal sheet of light (180 mm long, 30 mm wide, 1 mm high, wave length of 650 nm and output power of 3 mW). The particle passing through the light sheet causes a decrease of signal due to extinction. The amplitude of the signal deviation is a measure of particle size, and the duration of the signal allows an estimate of particle fall velocity. The size and fall velocity of each particle are sorted into 32 velocity classes ranging from 0.05 to 20 m/s and 32 particle size classes ranging from 0.062 to 24.5 mm. A detailed explanation about working, limitations and quality controlled procedures are explained by previous researchers [4-6].

The raindrop concentration $N(D_i)$ ($\text{mm}^{-1} \text{m}^{-3}$) at an instant of time from the PARSIVEL disdrometer counts is obtained from the following equation,

$$N(D_i) = \sum_{j=1}^{32} \frac{n_{ij}}{A_{\text{eff}} \Delta t V_j \Delta D_i} \text{ --- (1)}$$

Where n_{ij} is the number of drops reckoned in the size bin i and velocity bin j , Δt (s) is the sampling time, and ΔD_i (mm) is the width of the i^{th} class diameter, V_j (m/s) is the j^{th} class mean velocity and A_{eff} (m^2) is effective sampling area [7] given by $A_{\text{eff}} = 10^{-6} \times L \left(B - \frac{D_i}{2} \right)$ in which D_i (mm) is the i^{th} bin size drop diameter, L (mm) is the length of the parsivel beam (180 mm) and B (mm) is the width of the beam (30 mm).

From the raindrop concentration $N(D_i)$, drop diameter (D_i) and terminal fall velocity $V(D_i)$, the radar reflectivity factor Z ($\text{mm}^6 \text{m}^{-3}$) and rain rate R (mm/h) are derived by

$$Z(\text{dBZ}) = 10 \log_{10} \sum_{i=1}^{32} N(D_i) D_i^6 \Delta D_i \text{ --- (2)}$$

$$R(\text{mmh}^{-1}) = 6\pi \times 10^{-4} \sum_{i=1}^{32} N(D_i) D_i^3 V(D_i) \Delta D_i \text{ --- (3)}$$

The n^{th} order moment of the drop size distributions is expressed as

$$M_n = \int_{D_{\min}}^{D_{\max}} D^n N(D) dD \text{ --- (4)}$$

Where $n=3$ for 3rd moment, 4 for 4th moment and 6 for 6th moment of the size distribution.

The mass-weighted mean diameter (D_m in mm) is obtained from the 3rd and 4th moments of the size distribution as

$$D_m = \frac{M_4}{M_3} \text{ --- (5)}$$

The cloud effective radius (CER) values of ice and liquid particles are obtained from Moderate Resolution Imaging Spectroradiometer (MODIS) level 3 product. MODIS level 3 daily global atmospheric data product (MOD08_D3) consists of $1^\circ \times 1^\circ$ grid average values of atmospheric parameters related to aerosol particle properties, water vapor, cloud optical and physical properties [8]. The CER of liquid and ice phases are obtained from multispectral reflectance. The basic physical principle behind the retrieval of CER is the bispectral solar reflectance method first described by Nakajima and King (1990) [9] and applied to airborne data. Platnick et al. (2003) [10] and King et al. (2003) [11] provided detailed description about MODIS cloud product algorithms with example results.

3. Observational Results

The spatial variability of RSD in north east monsoon seasons of Kadapa (2010) and Gadanki (2011) locations are studied using parsivel disdrometer measurements. The rain drop concentration $[N(D)]$ variations with drop diameter for Kadapa and Gadanki rainfall is given in Figure 1. A clear demarcations of RSD concentration of Kadapa and Gadanki can be seen. The concentrations of mid ($1 \text{ mm} < \text{raindrop diameter} < 3 \text{ mm}$) and large (raindrop diameter $> 3 \text{ mm}$) drops are higher in Gadanki rainfall compared to Kadapa rainfall. Figure 2 depicts the frequency distribution of mass weighted mean diameter (D_m) of Kadapa and Gadanki rainfall. Both Kadapa and Gadanki rainfall showing positive skewness in mass weighted mean diameter distribution with higher mean value of D_m in Gadanki rainfall compared to Kadapa rainfall.

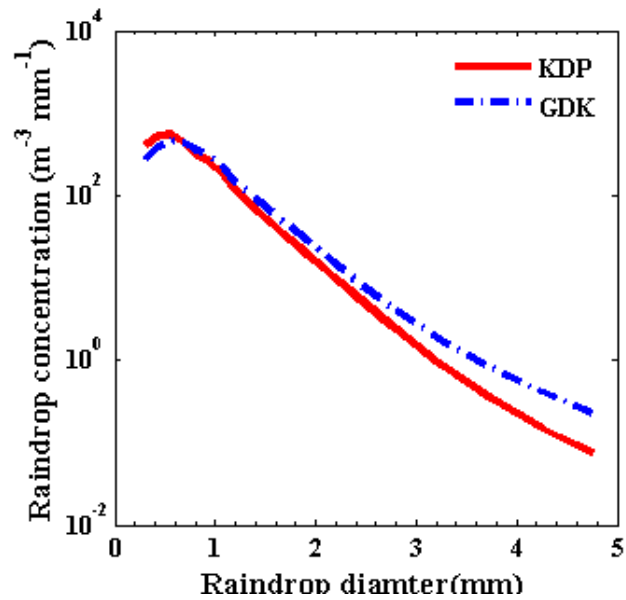


Figure 1. Variations in raindrop concentrations of Kadapa (KDP) and Gadanki (GDK) rainfall with drop diameter.

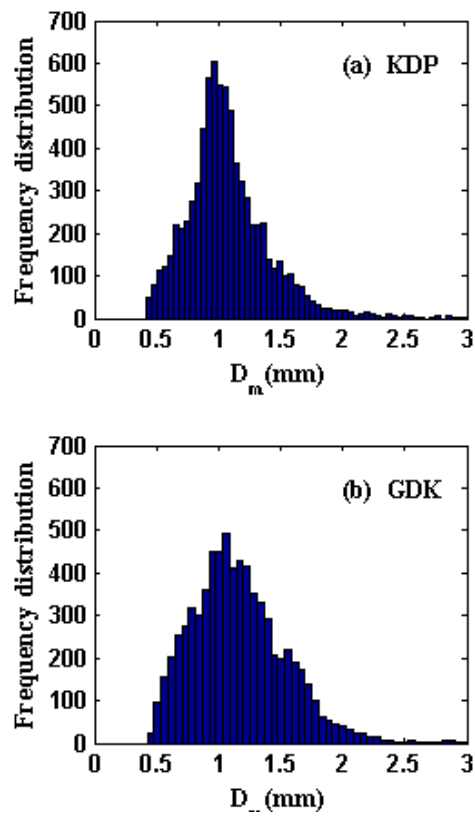


Figure 2. Frequency distribution of Mass weighted mean diameter (D_m) in (a) Kadapa (KDP) and (b) Gadanki (GDK) rainfall.

Monthly variation of D_m in Kadapa and Gadnki rainfall of north east monsoon is shown in Figure 3. The mean D_m values are higher in Gadnki rainfall compared to Kadapa rainfall in all the months (September to December). For both the locations mean D_m is showing a decreasing trend from September to December. This could be due to higher amount of convective rainfall in September and October months compared to November and December months (Rao et al., 2009). Diurnal variations in mass weighted mean diameter of Kadapa and Gadnki rainfall is provided in Figure 4. Gadnki is observed with higher D_m values in all the time interval (00-06, 06-12, 12-18, 18-24 hrs) than Kadapa rainfall.

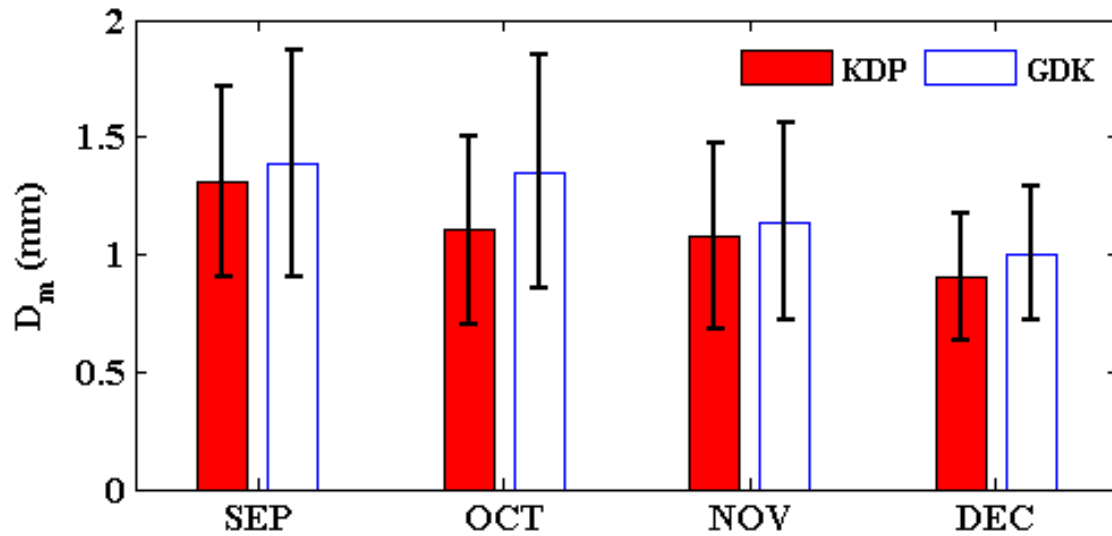


Figure 3. Monthly variation of mass weighted mean diameter (D_m) for Kadapa (KDP) and Gadnki (GDK) rainfall.

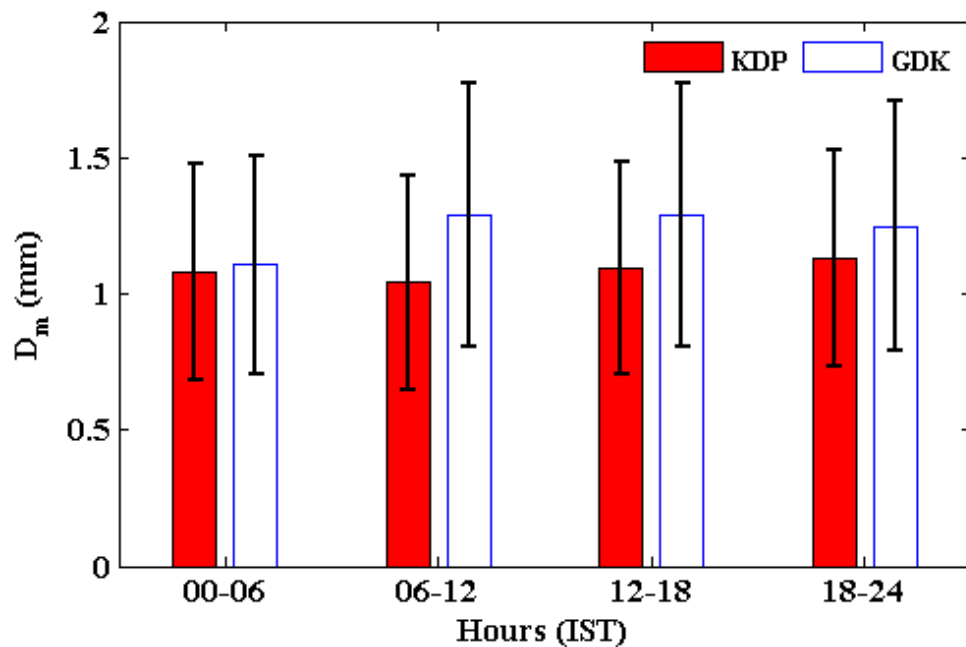


Figure 4. Diurnal variations (six hourly) of mass weighted mean diameter (D_m) of Kadapa (KDP) and Gadnki (GDK) rainfall.

Kadapa and Gadanki precipitations are classified into stratiform and convective regimes on the basis of rainfall rate criteria proposed by Bring et al., (2003). The mass weighted mean diameter is having higher values in stratiform precipitations of Gadanki than Kadapa and this feature is reversed for the convective precipitation. The cloud effective radius of liquid and ice particles obtained from the MODIS satellite for Kadapa and Gadanki locations are shown in Figure 6. Both ice and liquid phase particles have higher radii in Gadanki than Kadapa.

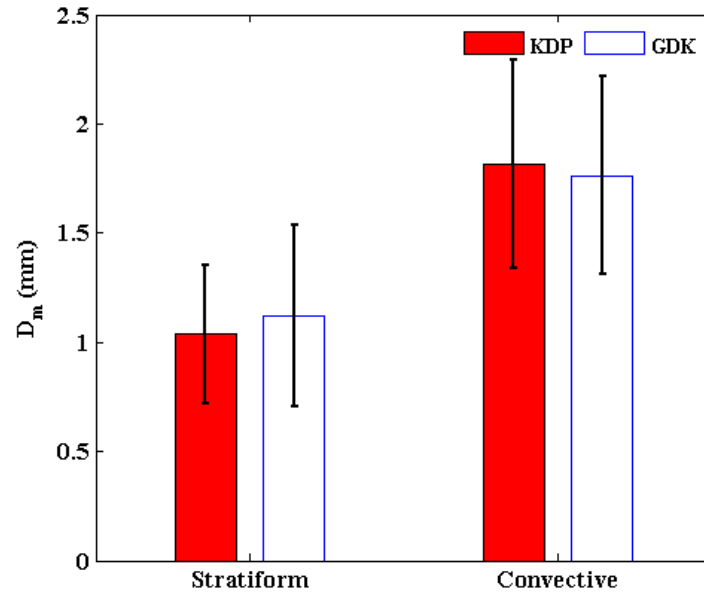


Figure 5. Variation of mass weighted mean diameter (D_m) in stratiform and convective precipitations of Kadapa (KDP) and Gadanki (GDK) rainfall.

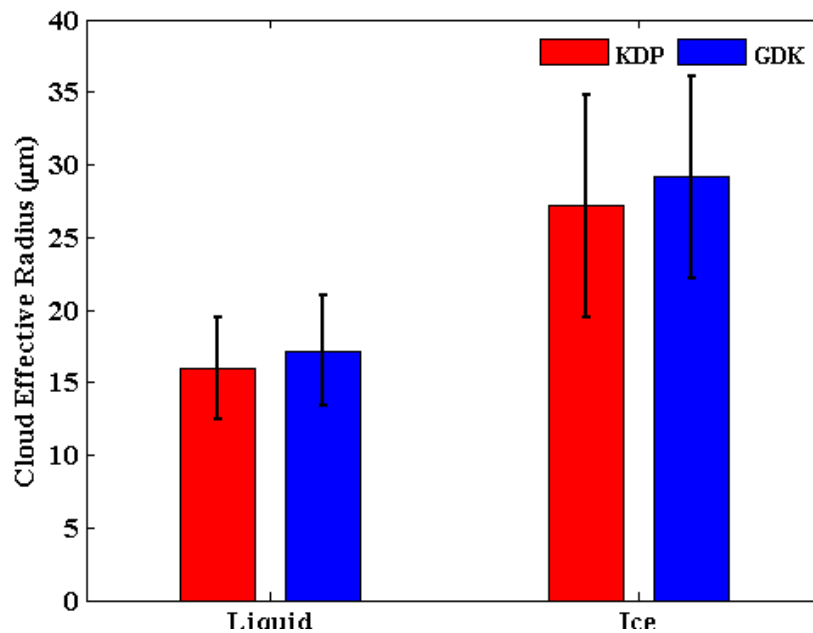


Figure 6. Cloud effective radius of liquid and ice particles of precipitating clouds observed by MODIS satellite for Kadapa (KDP) and Gadanki (GDK) locations.

4. Conclusions

Variations in raindrop size distribution (RSD) characteristics of two southern Indian stations (Kadapa, Gadanki) studied using Parsivel disdrometer data sets. Spatial variability of RSD between Kadapa and Gadanki showed a clear difference in mass weighted mean diameter between Gadanki and Kadapa. The raindrop concentration of mid and large drops is higher in Gadanki than Kadapa. Monthly and diurnal variations of D_m showed higher values in Gadanki than Kadapa rainfall. Higher cloud effective radii values (Figure 6) for Gadanki indicates the extent of clouds to higher altitudes with pronounced aggregation and riming process in the cloud and enhanced collision- coalescence process below the cloud [1] leading to higher D_m values in Gadanki rainfall than Kadapa rainfall.

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